Python Activity-Based Learning
Case Study

11 Examination Result and Grace-Mark Validation

Selected case study | Prepared using the topics listed in the supplied assignment. ## 1. Problem

Statement

Accept marks in multiple subjects, check subject-wise pass requirements, apply a limited grace-mark rule, and calculate the final result without exceeding maximum marks.

2. Objectives

• Store marks in a list.

• Check subject-wise pass criteria using conditions.

• Apply grace marks within the stated policy.

• Calculate total, average, and result.

3. Assumptions

• There are five subjects, each out of 100.

• The pass mark is 35 in every subject.

• A student short by at most 3 marks in exactly one subject may receive the required grace marks.

• Final marks cannot exceed 100.

4. Algorithm

1. Read marks for five subjects and validate that each is between 0 and 100.

2. Identify subjects below 35.

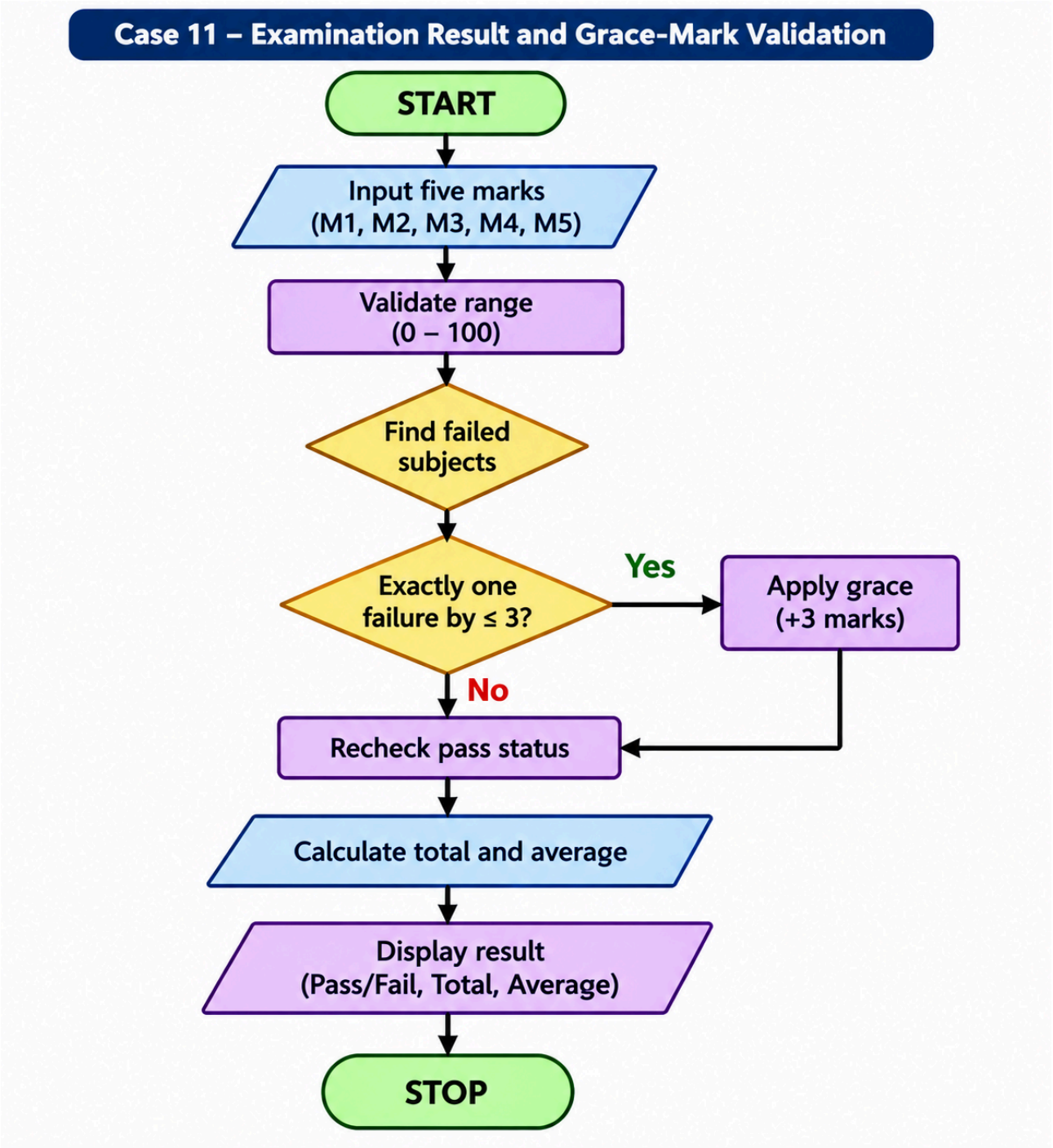
3. If exactly one subject is short by 1 to 3 marks, add enough grace marks to reach 35.

4. Check whether all subjects pass after grace marks.

5. Calculate total and average.

6. Display adjusted marks, grace marks, and result.

5. Flowchart



6. Pseudocode

```
BEGIN
 INPUT marks for 5 subjects
 Validate each mark is between 0 and 100
 Find marks below 35
 IF one subject is below 35 AND shortfall <= 3 THEN
 Add shortfall as grace marks
 END IF
 IF every mark >= 35 THEN result ← PASS
```

```
 ELSE result ← FAIL
 END IF
 PRINT marks, total, average, result
END
```

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7. Python Source Code

```
marks = []
for i in range(5):
 mark = int(input("Enter marks for subject " + str(i + 1) + ": "))
 if mark < 0 or mark > 100:
 print("Invalid mark. Enter again.")
 mark = int(input("Enter valid mark: "))
 marks.append(mark)

failed = []
for i in range(5):
 if marks[i] < 35:
 failed.append(i)

grace_used = 0
if len(failed) == 1:
 index = failed[0]
 shortfall = 35 - marks[index]
 if shortfall <= 3:
 marks[index] += shortfall
 grace_used = shortfall

total = sum(marks)
average = total / 5

if all(mark >= 35 for mark in marks):
 result = "PASS"
else:
 result = "FAIL"

print("\nAdjusted marks:", marks)
print("Grace marks used:", grace_used)
print("Total:", total)
print("Average:", average)
print("Result:", result)
```

8. Test Cases

Input / Scenario	Expected result
34, 60, 70, 80, 90	Grace 1; adjusted first subject 35; PASS
31, 60, 70, 80, 90	Grace not applied; FAIL
35, 40, 55, 65, 75	No grace; PASS

9. Conclusion

The program uses lists, loops, validation, and conditional logic to determine examination results under a defined grace-mark policy.

Note: Run the program in Python 3. The sample test cases are illustrative; enter the values when prompted.